

Task 2

1. Getting Started

Initialize a Git Repository: Create a new directory called **git-story**. Navigate into this directory and initialize a Git repository using `git init`

Basics

1. **Create chapter1.txt:** Write the following paragraph in a new file named **chapter1.txt**:

```
Once upon a time, in a village named "Init", lived a brave adventurer named Git.
```

Add the file to the staging area with `git add chapter1.txt` .

2. **Commit Your Changes:** Commit the changes with the message "Chapter 1: Introduce Git from Init village" using `git commit -m "Chapter 1: Introduce Git from Init village"` .
3. **Check the Status:** Use `git status` to check the status of your working directory and staging area.
4. **Show the Last Commit:** Use `git show` to display the changes made in the last commit.
5. **Check the Commit History:** Use `git log` to view the commit history. Try using `git log --oneline` to see a simplified one-line-per-commit version of the log.

Extend the Story

1. **Modify chapter1.txt:** Add more to the story in **chapter1.txt**. For example, add the following lines:

```
Git was well-known for his curiosity and eagerness to learn.  
He had one dream - to explore the mysterious world of Version Control.
```

Stage the changes with `git add chapter1.txt` .

2. **Commit the Changes:** Commit the changes with the message "Chapter 1: Extended story" using `git commit -m "Chapter 1: Extended story"` .
3. **Show and Log:** Use `git show` again to display the most recent commit. Then, use `git log` to see the new commit history.

By this point, you've used `git init` , `git add` , `git commit` , `git status` , `git show` , and `git log` in a realistic workflow.

2. Branching and Merging

1. **Create a New Branch:** Create a new branch called **chapter2** using `git branch chapter2`, and switch to it using `git checkout chapter2`.
2. **Create chapter2.txt:** Write a short paragraph about Git's decision to leave Init to explore the world of Version Control in **chapter2.txt**. For instance:

```
Chapter 2: Git's Decision
Git, after much contemplation, decided to leave his home in Init to
explore the vast world of Version Control.
```

Stage and commit the changes with the message "Chapter 2: Git's decision".

3. **Switch and Merge:** Switch back to the **master** branch using `git checkout master`. Merge the **chapter2** branch into **master** using `git merge chapter2`. This should be a fast-forward merge as there are no new commits on **master** that **chapter2** doesn't have.
4. **Non-Fast-Forward Merge:** Let's introduce a scenario where a fast-forward merge isn't possible. First, create and switch to a new branch **chapter3** and create **chapter3.txt** with content:

```
Chapter 3: Git's First Step into the World of Version Control
With a heavy heart and a bag full of hopes, Git took his first step
into the world unknown.
```

5. Commit this change. Now, switch back to **master**, modify **chapter1.txt** or **chapter2.txt**, and commit this change. Try merging **chapter3** into **master**. This will be a non-fast-forward merge as there have been new commits on both **master** and **chapter3**.
6. **Simulate Merge Conflicts:** To create a merge conflict scenario, let's create a new branch from **master** called **conflict_branch**. In this branch, change the first line of **chapter1.txt** to:

```
Once upon a time, in a town named "Init", lived an enthusiastic explorer named Git.
```

Commit the changes. Now switch back to master and change the same line to:

```
Once upon a time, in a village named "Init", lived a brave adventurer named Git.
```

Commit the changes here too. Now, try to merge **conflict_branch** into **master**. This should result in a merge conflict. Solve this conflict, decide which version to keep, stage, and commit the merge.

By the end of this part, you would have practiced creating branches (**git branch**), switching branches (**git checkout**), merging branches (**git merge**), resolving conflicts, and experienced both fast-forward and non-fast-forward merges.

